

Linux Fundamentals Lab Exercise

Objective

This lab demonstrates basic Linux commands for file management, text processing, shell scripting, and environment variables. Each task below lists the original question, the exact command(s) executed, and a terminal screenshot showing the actual output.

Task 1: Directory and File Operations

Question: Create a directory named `linux_lab`, navigate into it, create the files `students.txt`, `script.sh` and `output.txt`, then display the current working directory and list all files.

Command(s) used: `mkdir linux_lab → cd linux_lab → touch students.txt script.sh output.txt → pwd → ls -l`

Result / Screenshot:

```
sabarish@Jarvis:~$ mkdir linux_lab
sabarish@Jarvis:~$ cd linux_lab
sabarish@Jarvis:~/linux_lab$ touch students.txt script.sh output.txt
sabarish@Jarvis:~/linux_lab$ pwd
/home/sabarish/linux_lab
sabarish@Jarvis:~/linux_lab$ ls -l
total 0
-rw-rw-r-- 1 sabarish sabarish 0 Aug  5 20:55 output.txt
-rw-rw-r-- 1 sabarish sabarish 0 Aug  5 20:55 script.sh
-rw-rw-r-- 1 sabarish sabarish 0 Aug  5 20:55 students.txt
sabarish@Jarvis:~/linux_lab$
```

Result: The directory `linux_lab` was created and entered. All three required files (`students.txt`, `script.sh`, `output.txt`) were created successfully, as confirmed by the `pwd` and `ls -l` output above.

Task 2: Create and Edit a File

Question: Using a Linux text editor (`nano` or `vim`), add the given student records to `students.txt`, save it, and verify its contents.

Command(s) used: `nano students.txt (edit & save) → cat students.txt`

Result / Screenshot:

```
GNU nano 8.7.1 students.txt *
Alice, Python
Bob, Docker
Charlie, Linux
David, Python
Eva, Docker
Frank, Linux
Grace, Python
Henry, Docker
```

Result: `students.txt` was edited in `nano` and saved with all 8 student records. The `cat` command confirms the file contents match the required data exactly.

Task 3: Text Processing

Question: Display all records containing "Python", "Docker", and "Linux" respectively, using grep.

Command(s) used: `grep "Python" students.txt` | `grep "Docker" students.txt` | `grep "Linux" students.txt`

Result / Screenshot:

```
sabarish@Jarvis:~/linux_lab$ grep "Python" students.txt
Alice, Python
David, Python
Grace, Python
sabarish@Jarvis:~/linux_lab$ grep "Docker" students.txt
Bob, Docker
Eva, Docker
Henry, Docker
sabarish@Jarvis:~/linux_lab$ grep "Linux" students.txt
Charlie, Linux
Frank, Linux
sabarish@Jarvis:~/linux_lab$
```

Result: grep successfully filtered and displayed 3 Python records, 3 Docker records, and 2 Linux records from students.txt.

Question: Count the number of records containing "Python", save that count into output.txt, and display the contents of output.txt.

Command(s) used: `grep -c "Python" students.txt` → `grep -c "Python" students.txt > output.txt` → `cat output.txt`

Result / Screenshot:

```
sabarish@Jarvis:~/linux_lab$ grep -c "Python" students.txt
3
sabarish@Jarvis:~/linux_lab$ grep -c "Python" students.txt > output.txt
sabarish@Jarvis:~/linux_lab$ cat output.txt
3
sabarish@Jarvis:~/linux_lab$
```

Result: The count of Python records is 3. This value was redirected into output.txt using the > operator, and cat output.txt confirms it was saved correctly.

Task 4: Shell Scripting

Question: Create a shell script script.sh that displays a welcome message, the current user, the current working directory, and the current date & time. Run the script from the terminal.

Command(s) used: `nano script.sh` (write script) → `cat script.sh` → `chmod +x script.sh` → `./script.sh`

Result / Screenshot:

```
sabarish@Jarvis:~/linux_lab$ cat script.sh
#!/bin/bash
echo "Welcome to Linux Lab"
echo "Current User: $(whoami)"
echo "Current Working Directory: $(pwd)"
echo "Current Date and Time: $(date)"
sabarish@Jarvis:~/linux_lab$ chmod +x script.sh
sabarish@Jarvis:~/linux_lab$ ./script.sh
Welcome to Linux Lab
Current User: sabarish
Current Working Directory: /home/sabarish/linux_lab
Current Date and Time: Wed Aug 5 09:06:26 PM IST 2026
sabarish@Jarvis:~/linux_lab$
```

Result: script.sh was created with the required echo statements using \$(whoami), \$(pwd) and \$(date), made executable with chmod +x, and executed successfully — printing the welcome message, current user, working directory, and system date/time.

Task 5: Environment Variables

Question: Create an environment variable `COURSE_NAME` with the value "Linux Fundamentals", display its value, verify it is exported, and then modify script.sh to also display `COURSE_NAME` along with the other system information.

Command(s) used: `export COURSE_NAME="Linux Fundamentals" → echo $COURSE_NAME → env | grep COURSE_NAME → nano script.sh (edit) → cat script.sh → ./script.sh`

Result / Screenshot:

```
sabarish@Jarvis:~/linux_lab$ export COURSE_NAME="Linux Fundamentals"
sabarish@Jarvis:~/linux_lab$ echo $COURSE_NAME
Linux Fundamentals
sabarish@Jarvis:~/linux_lab$ env | grep COURSE_NAME
COURSE_NAME=Linux Fundamentals
sabarish@Jarvis:~/linux_lab$ cat script.sh
#!/bin/bash
echo "Welcome to Linux Lab"
echo "Course Name: $COURSE_NAME"
echo "Current User: $(whoami)"
echo "Current Working Directory: $(pwd)"
echo "Current Date and Time: $(date)"
sabarish@Jarvis:~/linux_lab$ ./script.sh
Welcome to Linux Lab
Course Name: Linux Fundamentals
Current User: sabarish
Current Working Directory: /home/sabarish/linux_lab
Current Date and Time: Wed Aug 5 09:09:34 PM IST 2026
sabarish@Jarvis:~/linux_lab$
```

Result: The `COURSE_NAME` environment variable was created and exported successfully — confirmed via `echo` and `env | grep COURSE_NAME`. script.sh was updated to include an extra echo line for `$COURSE_NAME`, and running the updated script confirms it now prints the course name alongside the user, directory, and date/time.

Final File Contents (Submission)

students.txt

```
Alice,Python
Bob,Docker
Charlie,Linux
David,Python
Eva,Docker
Frank,Linux
Grace,Python
Henry,Docker
```

script.sh

```
#!/bin/bash
echo "Welcome to Linux Lab"
echo "Course Name: $COURSE_NAME"
echo "Current User: $(whoami)"
echo "Current Working Directory: $(pwd)"
echo "Current Date and Time: $(date)"
```

output.txt

```
3
```

Conclusion

All five tasks were completed successfully. This lab reinforced practical skills in directory/file operations (mkdir, touch, pwd, ls), file editing and verification (nano, cat), text processing and filtering (grep, grep -c, output redirection), shell scripting with command substitution (\$(...)), and creating/exporting/using environment variables (export, echo, env) inside a shell script.